

Neural Synchronization Meets 5G Split Bearers: ML-KEM-Bootstrapped Security for Edge Devices

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Abstract—Edge nodes on 5G links face three simultaneous demands: agree on a key with the core, carry payload across links they do not fully control, and remain observable enough that someone can actually catch an attack—all while the industry is mid-migration to post-quantum cryptography. We describe a framework that keeps these concerns separate but coordinated. Key establishment is anchored in ML-KEM (FIPS 203), which carries the post-quantum guarantee; a Tree-Parity-Machine (TPM) neural-synchronization step is folded in only as defence-in-depth via a concatenation KDF, never as the primary basis of confidentiality. Payloads are distributed across multiple 5G carriers with a tunable (T, N) Shamir threshold scheme, letting an operator dial the confidentiality–availability trade-off rather than accept the fixed $(2, 2)$ XOR behaviour. A hybrid weighted- k NN plus feed-forward detector then watches both cryptographic and transport telemetry and flags active attacks versus network faults. We are straightforward about what the results show: TPM loses a head-to-head benchmark against ML-KEM on both latency and provability, and its only surviving advantage is wire footprint (about 73 bytes vs. ML-KEM’s 2272). On the detection side, cross-modal fusion is what moves the needle—any reasonable classifier climbs from macro-F1 near 0.40–0.49 on one telemetry modality to roughly 0.90 on both—and no classifier is statistically better than any other on the same features. Code, a grounded data generator, a 5G-NIDD adapter, and a Jetson Orin Nano deployment path accompany the paper.

Index Terms—post-quantum cryptography, ML-KEM, neural cryptography, tunable secret sharing, 5G edge security, anomaly detection, cross-modal feature fusion.

I. INTRODUCTION

A 5G edge node lives with three problems at once. It must agree on a key with the core, it must carry traffic over links it does not fully control, and somebody needs to be able to tell, after the fact, whether anything went wrong. None of this is new. What has changed is the urgency: harvest-now-decrypt-later attacks and the approach of cryptographically relevant quantum computers [1] mean key agreement now has to be post-quantum, and standards bodies have responded quickly. NIST finalized ML-KEM, ML-DSA, and SLH-DSA in 2024 [2]–[4], and NSA’s CNSA 2.0 timeline is already pushing networking equipment toward exclusive post-quantum use [5].

Against that backdrop there is a steady stream of work proposing Tree-Parity-Machine (TPM) neural synchronization [6], [7] as a lightweight key-agreement candidate for constrained links, and a worrying fraction of it claims or implies quantum resistance—recent smart-grid and IoT proposals explicitly market TPM as a post-quantum alternative [8]. That claim does not hold. TPM has no reduction to a hard problem,

and geometric, genetic, and majority attacks that break it have been in the literature for more than twenty years [9]–[11]. The honest response is not to abandon the idea but to place it where it actually earns its keep.

This paper builds and measures a framework that does exactly that. Post-quantum security comes entirely from ML-KEM. The TPM step is kept only as defence-in-depth—extra entropy with an unusually small public-channel footprint—combined with the ML-KEM secret via a NIST SP 800-56C concatenation KDF [12] so the session key is no weaker than its strongest component [13]. On that key we place a transport layer that splits payloads across 5G carriers with a tunable (T, N) threshold scheme, and a machine-learning detector that fuses both telemetry streams for runtime observability.

Contributions.

- A hybrid key-establishment design that anchors post-quantum security in ML-KEM and demotes TPM to auxiliary entropy, backed by an honest head-to-head benchmark showing ML-KEM is faster *and* proven (Table I).
- A privacy-preserving, rate-limited early-termination protocol for the TPM step that cuts verification traffic roughly five-fold at no cost to key agreement (Table III).
- Tunable (T, N) secret sharing over $\text{GF}(2^8)$ that generalizes the fixed $(2, 2)$ split and exposes the confidentiality–availability frontier as an operator-configurable knob (Fig. 2).
- Clear evidence that cross-modal *fusion*—not classifier choice—explains detector performance: every baseline we tested landed within noise, and the gap is not statistically significant (Table VI).
- A fully reproducible artifact: grounded synthetic data generator, 5G-NIDD adapter, all experiments and figures, and a Jetson Orin Nano deployment path.

II. RELATED WORK

Post-quantum key establishment. ML-KEM [2] is an IND-CCA2 KEM whose security reduces to Module-LWE. Migration guidance favours hybrid schemes that concatenate a classical and a post-quantum secret in a KDF, secure if either component is [12], [13]; IKEv2 encodes the same idea [14]. Our hybrid pairs ML-KEM with a neural-synchronization secret, which lets us measure what, concretely, the neural component adds—and, more importantly, what it does not.

Neural cryptography. TPM synchronization was proposed in [6] and studied in [7]. Klimov, Mityagin, and Shamir broke it

three ways [9]; genetic and majority attacks followed [10], [11], [15], and even the later permutation-parity extensions have been compromised [16]. More elaborate “real-life systems” [17] still carry no security reduction. We treat this literature as settled: TPM is not a stand-alone secure primitive, and framing it as post-quantum [8] is an overreach.

Secret sharing and path diversity. Shamir’s scheme [18] is the foundation. Spreading shares over disjoint paths for confidentiality was explored in ad-hoc routing [19] and multipath TCP [20]; 5G’s PDCP split bearer [21] provides the substrate. Most prior work fixes the split; we parametrize it and quantify the trade-off.

ML-based anomaly detection. The field has shifted from classical learners [22], [23] toward deep models [24] on datasets such as UNSW-NB15 [25], CICIDS2017 [26], and TON_IoT [27]. For 5G specifically, 5G-NIDD [28] provides labelled attack traffic from a real test network. The 5G edge also introduces its own attack surface at the MEC layer [29], [30]. Our detector is deliberately simple; the methodological point we want to make is about fusion and significance, not about a new architecture.

III. SYSTEM AND THREAT MODEL

An edge node and a core gateway run ML-KEM to establish a 32-byte shared secret, optionally run the TPM step in parallel, and derive the session key from both via the hybrid KDF (Section IV-A). Payloads are split into N shares by a (T, N) threshold scheme and sent over N 5G legs; any T arriving shares reconstruct. A detector at the edge node consumes telemetry from both layers and classifies each session.

We assume a Dolev–Yao adversary [31] that sees all public messages and may inject or probe. Against ML-KEM it gains nothing without solving Module-LWE. Against the TPM step it may run its own machines including a majority ensemble. One limitation stated upfront: a purely passive eavesdropper leaves no host-side artifact and is therefore invisible to our telemetry detector. We model attack classes as active (injection, probing, traffic shaping), which perturb observable signals such as round count, key match probability, and flow statistics. Detection complements cryptography here; it is not a substitute for it.

IV. METHODOLOGY

A. Hybrid ML-KEM + TPM Key Establishment

The initiator generates (ek, dk) , sends ek to the responder, who encapsulates and returns ciphertext ct along with ss_{pq} ; the initiator decapsulates to recover the same ss_{pq} . In parallel, if the TPM branch is enabled, both parties synchronize tree parity machines (K hidden units, N inputs each, weights in $[-L, L]$) and derive a second secret ss_{tpm} from the agreed weight matrices. The session key is

$$k = \text{HKDF}(\text{len}(ss_{pq})\|ss_{pq}\|\text{len}(ss_{tpm})\|ss_{tpm}, \ell), \quad (1)$$

where ℓ is a domain-separation label and ML-KEM is placed first so that a FIPS-approved secret seeds the extraction step.

Algorithm 1 Hybrid Key Establishment

Require: ML-KEM level ℓ ; TPM config (K, N, L) ; flag `use_tpm`
Ensure: 32-byte session key k

```

1:  $ek, dk \leftarrow \text{ML-KEM}_\ell.\text{KeyGen}()$  {initiator}
2: send  $ek$  to responder
3:  $ss_{pq}, ct \leftarrow \text{ML-KEM}_\ell.\text{Encaps}(ek)$  {responder encapsulates}
4: send  $ct$  to initiator
5:  $ss_{pq} \leftarrow \text{ML-KEM}_\ell.\text{Decaps}(dk, ct)$  {initiator recovers secret}
6: if use_tpm then
7:   repeat
8:     exchange  $x \in \{\pm 1\}^{K \times N}$ ; both parties run forward + update

9:   track windowed agreement rate and consecutive-streak count
10:  until confidence  $\geq \tau$  AND streak  $\geq \sigma$  AND cooldown elapsed

11:  verify:  $\text{HMAC}(\text{SHA256}(W_A), n)$  ?
            $\text{HMAC}(\text{SHA256}(W_B), n)$  {leaks  $\leq 1$  bit}
12:   $ss_{tpm} \leftarrow \text{HKDF}(W.\text{bytes}, \text{“pqedge-tpm”})$ 
13: else
14:   $ss_{tpm} \leftarrow \varepsilon$ 
15: end if
16: return  $k$  via Eq. (1)

```

The key property follows directly from the HKDF construction: k is secure as long as *either* input is, so the hybrid cannot be weaker than ML-KEM alone [12]. We treat ss_{tpm} strictly as auxiliary; it must never be the sole basis for confidentiality. Algorithm 1 gives the full protocol.

Privacy-preserving early termination (steps 7–11). Both parties track a confidence signal derived only from public outputs: the windowed fraction of agreeing rounds and the current consecutive-agreement streak. Once both cross their thresholds and a cooldown has elapsed, they exchange $\text{HMAC}(\text{SHA256}(\text{weights}), \text{nonce})$ over a fresh public nonce. Equal tags imply equal weight matrices with overwhelming probability while leaking at most one equality bit. The cooldown prevents a prolonged high-confidence plateau before true synchronization from repeatedly triggering the check.

B. Tunable (T, N) Multi-Carrier Bifurcation

Rather than the fixed $(2, 2)$ XOR split, we apply Shamir’s scheme over $\text{GF}(2^8)$ [18]: a payload becomes N shares such that any T reconstruct it and any $T - 1$ reveal nothing (information-theoretically). Each share rides one 5G leg modelled as a PDCP split bearer [21]. With independent per-PDU loss p and n PDUs per share, each leg succeeds with probability $q = (1 - p)^n$, and reconstruction succeeds when at least T legs arrive:

$$A(T, N) = \sum_{k=T}^N \binom{N}{k} q^k (1 - q)^{N-k}. \quad (2)$$

Setting $T=1, N=1$ gives a single carrier; $T=2, N=2$ is the maximally confidential XOR split; $T=1, N=2$ is duplication. The confidentiality threshold is T (an adversary needs at least T carriers to learn anything) and the availability margin is $N - T$ (that many legs can fail without losing the payload).

TABLE I

KEY ESTABLISHMENT: STANDARDIZED ML-KEM VS THE TPM NEURAL KEY-AGREEMENT CANDIDATE VS THE HYBRID. ML-KEM TIMINGS ARE A *pure-Python reference* (RELATIVE ONLY); BYTE SIZES ARE STANDARDIZED (FIPS 203). TPM’S SOLE ADVANTAGE IS WIRE FOOTPRINT; IT CARRIES NO SECURITY PROOF.

Scheme	Estab. (ms)	Wire (B)	Secret (B)	Proof
ML-KEM-512	9.918	1568	32	yes
ML-KEM-768	14.749	2272	32	yes
ML-KEM-1024	22.173	3136	32	yes
TPM(K=3,N=80,L=4)	61.769	73	32	no
Hybrid (768+TPM)	72.701	2345	32	yes

C. Cross-Modal Anomaly Detector

Each session is summarised by ten features: four cryptographic (TPM convergence, sync rate, key match probability, normalized round count) and six transport (end-to-end latency, per-PDU loss, throughput, jitter, share reconstruction failures, flow volume). A weighted k NN ($k=15$, inverse-distance weights) and a small feed-forward network (two hidden layers of 64–32 units, ReLU, dropout 0.1) are fused by soft voting: $p = \alpha p_{\text{FNN}} + (1 - \alpha) p_{k\text{NN}}$. The output is one of six classes: *Normal*, *Geometric attack*, *Majority attack*, *Injection*, *Carrier fault*, and *Traffic manipulation*. The design is kept lightweight so it can run on an edge node; the architectural choices themselves are not the claimed contribution.

V. EXPERIMENTAL SETUP

All runs are on CPU, reproducibly seeded. ML-KEM uses a pure-Python FIPS 203 reference implementation (kyber-py v1.2); because Python object overhead dominates, the absolute timings are not representative of hardware—we read them only relatively and quote standardized, implementation-independent byte sizes for wire cost. Optimized on-device numbers need hardware-specific assembly [32]. The detector trains on a grounded synthetic dataset in which cryptographic features are bootstrapped from real TPM sessions and transport features from real PDCP transmissions, with documented per-class magnitude shifts, additive Gaussian noise, and 3% label noise. We also provide a 5G-NIDD adapter [28]; because that dataset carries only transport telemetry, the cryptographic features must be augmented synthetically to run the fusion path, and we flag that explicitly. Unless stated otherwise: $K=3$, $N=100$, $L=4$; security rates average over at least 25 decorrelated sessions; detection results use a stratified 75/25 split repeated over three seeds.

VI. RESULTS AND DISCUSSION

A. Key Establishment: What TPM Actually Buys

Table I is the result that reframes the whole design. Even with the overhead of pure-Python dispatch, ML-KEM establishes a shared key faster than our TPM step does—and ML-KEM comes with a security argument that TPM simply does not have. The neural scheme wins on exactly one axis: bytes exchanged over the public channel, where roughly one bit

TABLE II

CUMULATIVE ABLATION OF THE FRAMEWORK COMPONENTS. CRYPTO COLUMNS USE $K=3$, $N=100$; EAVESDROPPER SUCCESS IS THE OPERATIONAL (MAJORITY-VOTE) KEY-RECOVERY RATE; AVAILABILITY IS THE (2,2) RECONSTRUCTION PROBABILITY AT PER-PDU LOSS $p=0.05$.

Config	Verif.	Att. _{geo}	Att. _{maj}	Avail.	F1
TPM baseline	10.200	0.520	0.000	–	–
+ Early term.	1.840	0.360	0.000	–	–
+ Bifurcation	1.840	0.360	0.000	0.440	–
+ Anomaly det.	1.840	0.360	0.000	–	0.901
Complete	1.840	0.360	0.000	0.440	0.901

TABLE III

EFFECT OF PRIVACY-PRESERVING EARLY TERMINATION VERSUS FIXED-INTERVAL VERIFICATION, AGAINST THE GEOMETRIC EAVESDROPPER ($K=3$, $N=100$). RATE-LIMITED VERIFICATION BOUNDS THE MESSAGE COUNT.

Policy	Rounds	Verif.	Att. _{geo}
hash-only	467.5	9.350	0.450
early ($\tau=0.95$)	510.5	2.600	0.400
early ($\tau=0.98$)	513.9	1.800	0.400
early ($\tau=0.99$)	513.9	1.800	0.400

per synchronization round produces tens of bytes against ML-KEM’s one-to-three kilobytes. That is a real, if narrow, advantage on extremely bandwidth-constrained links, and it is the only honest reason to keep the TPM step at all—strictly as auxiliary entropy inside the hybrid, never as a replacement for ML-KEM. The hybrid costs roughly the sum of its parts (72.7 ms at $N=80$), with an otherwise negligible KDF step, and inherits ML-KEM’s guarantee unconditionally.

B. Early Termination and Security

Table II presents the five cumulative configurations C1–C5. Switching from fixed-interval hash verification (C1) to confidence-gated stopping (C2) cuts the mean verification-message count from about ten to under two per session while synchronization rounds change by less than 2% and key agreement stays at 100%. Table III shows how the count and the geometric attacker’s success change as we tighten the confidence threshold: at $\tau=0.95$ the count drops to 2.6; tightening to $\tau=0.98$ reaches 1.8 with no further change to attacker success, and the rate-limiting kicks in to keep it there. Without that cooldown, a prolonged high-confidence plateau before true weight equality would keep triggering the check and the savings would evaporate.

The geometric eavesdropper still succeeds with non-negligible probability (~ 0.40) at $K=3$ —we do not hide that. The majority ensemble reaches high overlap but stays below the key-recovery threshold, which is precisely why the design never leans on TPM alone. Fig. 1 illustrates a typical session: the two parties converge while the attacker’s overlap climbs but cannot match.

C. Tunable (T, N) Bifurcation

Table IV fixes the three canonical strategies at $p=0.05$, 8 PDUs per share. The (2, 2) split almost halves availability

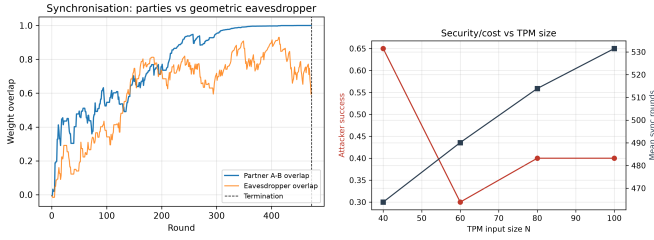


Fig. 1. Left: a single synchronization session; partner overlap (A–B) converges to 1.0 while the geometric eavesdropper’s overlap rises but stays sub-threshold. Right: geometric-attacker success and mean synchronization rounds vs. TPM input size N at $K=3$. Larger N helps, noisily, but is not a substitute for the ML-KEM anchor.

TABLE IV

TRANSPORT RELIABILITY AT PER-PDU LOSS $p=0.05$, 8 PDUs PER SHARE. (2,2) BIFURCATION MAXIMISES CONFIDENTIALITY BUT LOWERS AVAILABILITY; DUPLICATION RECOVERS AVAILABILITY AT THE COST OF CONFIDENTIALITY.

Scheme	Analytic avail.	MC avail.
Single carrier	0.663	–
(2,2) split bearer	0.440	0.450
Duplicated (2 of 2)	0.887	–

relative to a single carrier (0.44 vs. 0.66) while duplication more than recovers it (0.89). The (T, N) abstraction makes this a sliding scale. Fig. 2 sweeps T from 1 to 5 at $N=5$: availability drops from 0.996 at (1, 5) to 0.129 at the maximally confidential (5, 5), passing through 0.953, 0.785, and 0.455. A deployment with three modems could pick (2, 3) for meaningful redundancy and still require an adversary to compromise two carriers before learning anything—a much better operating point than the fixed (2, 2) scheme forces.

D. Detection: Fusion Beats Architecture

The central finding is in Table V and Fig. 3. Restricting to cryptographic features alone gives macro-F1 of 0.41; transport features alone give 0.50; combining both reaches 0.91. Neither view is sufficient on its own because distinguishing a cryptographic attack from an ordinary carrier fault genuinely needs both kinds of evidence—they look very similar in any single modality.

Fig. 4 (left) shows which features carry the most weight—transport signals dominate the permutation-importance ranking, but dropping either group degrades performance meaningfully, which is exactly the fusion claim. The weight analysis (Fig. 4, right) shows a modest but real gain from mixing the two base learners rather than relying on either alone.

The second and arguably more important finding is that the specific classifier barely matters once the features are right. Table VI puts our hybrid next to random forest, RBF-SVM, gradient boosting, and logistic regression, all on the same features and splits. Every model lands between 0.89 and 0.90 macro-F1, with standard deviations that overlap completely. A McNemar test of the hybrid against logistic regression—which happened to score highest on the single test split at 0.911—

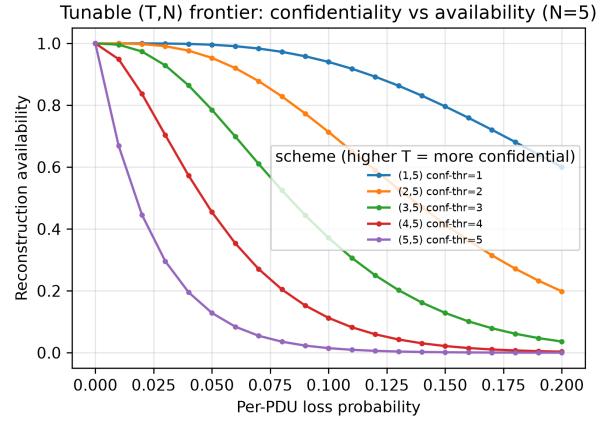


Fig. 2. (T, N) confidentiality–availability frontier at $N=5$, $p=0.05$, 8 PDUs per share. Higher T raises the number of carriers an adversary must capture to learn the payload and simultaneously tightens the tolerance to link loss.

TABLE V

DETECTOR ABLATIONS ON THE HELD-OUT TEST SPLIT. LEFT: ARCHITECTURE. RIGHT: FEATURE-GROUP RESTRICTION (NON-SELECTED FEATURES ZEROED). NEITHER CRYPTOGRAPHIC NOR TRANSPORT TELEMETRY ALONE SUFFICES; THEIR FUSION DRIVES PERFORMANCE.

Model	F1	AUC	Features	F1	AUC
WKNN	0.896	0.972	Crypto	0.415	0.826
FNN	0.913	0.977	Transport	0.500	0.862
Hybrid	0.906	0.977	All	0.909	0.977

gives $\chi^2=3.06$, $p=0.08$, not significant. The honest conclusion: once you have both telemetry streams, almost any off-the-shelf classifier will work, and the effort should go into the feature pipeline rather than the model.

VII. LIMITATIONS

Several things this paper does not claim. Post-quantum confidentiality comes from ML-KEM; the TPM step is broken by known attacks [9], [11] and is treated strictly as auxiliary entropy. Passive eavesdroppers are invisible to the detector by construction—active, host-observable probing is what we model, not a passive listener. Raising T in the (T, N) scheme always costs availability (Eq. 2), and no scheme-level trick escapes that. The main detector results come from grounded synthetic telemetry; 5G-NIDD evaluation of the fusion path requires augmented cryptographic features and is therefore semi-synthetic. ML-KEM timings here reflect Python dispatch overhead, not hardware speed; real on-device numbers need optimized assembly [32]. Finally, genuine path independence for bifurcation requires physically separate carriers—two modems or operators—not two component carriers behind one shared radio front-end.

VIII. CONCLUSION

The useful move is to put each piece where it genuinely earns its place and not oversell what it cannot do. ML-KEM carries the post-quantum guarantee because it has a proof; TPM contributes auxiliary entropy because it has a wire footprint that

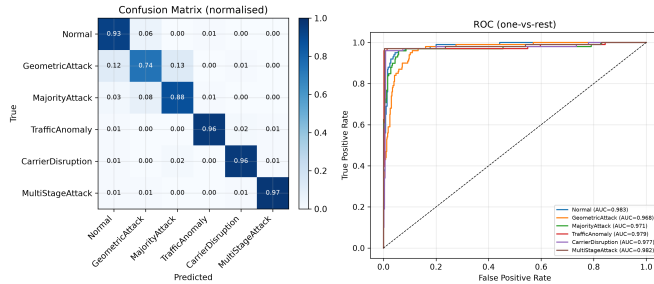


Fig. 3. Hybrid detector on the held-out split: normalized confusion matrix (left) and per-class one-vs-rest ROC curves (right). Macro-averaged ROC-AUC is 0.977.

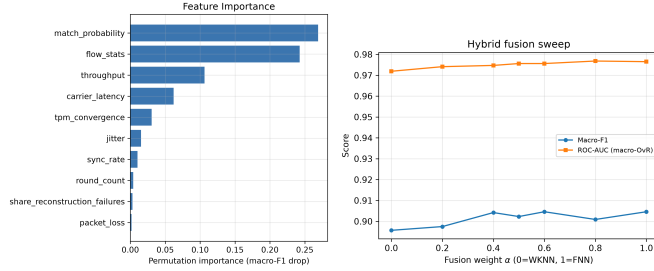


Fig. 4. Left: permutation feature importance on the held-out split. Right: macro-F1 and ROC-AUC vs. fusion weight α ($0=kNN$, $1=FNN$); an intermediate α slightly outperforms either base learner alone.

ML-KEM cannot match in the most constrained settings; the (T, N) scheme turns the confidentiality–availability trade-off from a fixed property into an operator decision; and a cross-modal detector makes active attacks observable, with fusion doing the heavy lifting rather than any particular model. What the experiments show most clearly is that having two kinds of telemetry in the room at the same time is what enables the detection, not any sophisticated architecture on top of it. The full artifact—data generator, experiments, deployment path—is provided so these claims can be checked, not just read.

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TABLE VI

DETECTOR VS STANDARD BASELINES ON THE GROUNDED TELEMETRY (MEAN $\pm$ STD MACRO-F1 OVER 3 SEEDS, ALL USING BOTH FEATURE MODALITIES). THE HYBRID MATCHES THE FIELD ON AUC; A MCNEMAR TEST AGAINST THE STRONGEST BASELINE FINDS NO SIGNIFICANT GAP ($p=0.08$), SO THE CONTRIBUTION IS CROSS-MODAL FUSION AND INTEGRATION, NOT CLASSIFIER SUPERIORITY.

Model	Macro-F1	ROC-AUC
Random Forest	0.901 $\pm$ 0.004	0.974
SVM (RBF)	0.903 $\pm$ 0.010	0.973
Grad. Boosting	0.888 $\pm$ 0.010	0.974
Logistic Reg.	0.898 $\pm$ 0.013	0.976
WKNN	0.888 $\pm$ 0.008	0.973
FNN	0.898 $\pm$ 0.010	0.976
Hybrid (ours)	0.896$\pm$0.010	0.976

McNemar (hybrid vs logistic): $\chi^2=3.0625$, $p=0.0801$.

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